

**BEFORE YOU READ — ANSWER FROM MEMORY**

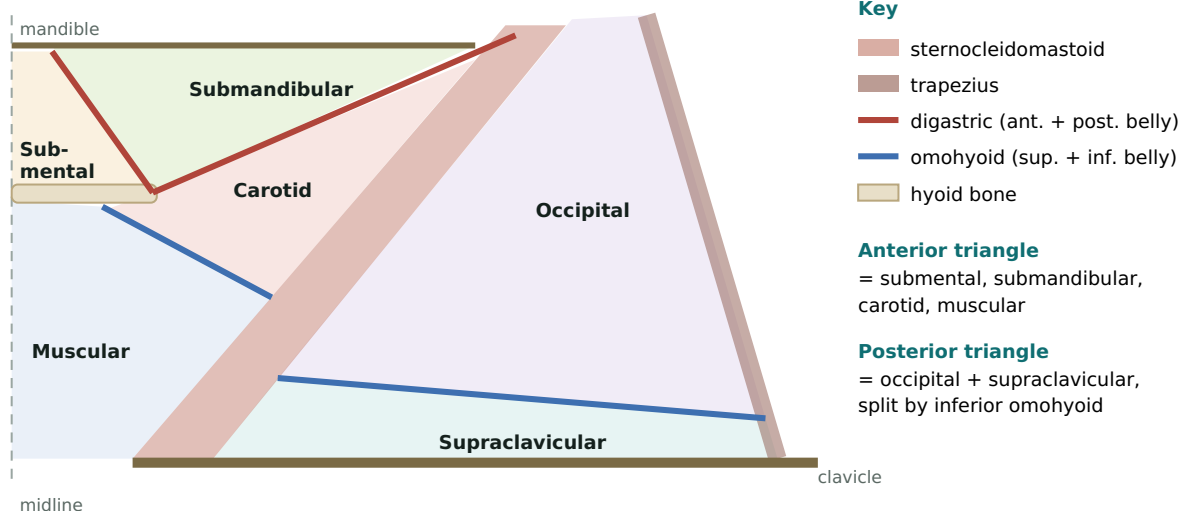
1. Name the three borders of the anterior and of the posterior triangle.
2. Which nerves lie in the carotid triangle?
3. Which structures pierce the investing layer of deep cervical fascia?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The two big triangles

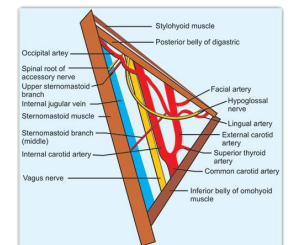
Two triangular areas in front of and behind **sternocleidomastoid** that contain the visceral structures of the neck.

	Anterior triangle	Posterior triangle
Borders	Above: inferior border of mandible · medial: midline of neck · lateral: anterior border of SCM	Front: posterior border of SCM · back: anterior border of trapezius · base: middle ⅓ of clavicle
Subdivisions	Submental · submandibular (digastric) · carotid · muscular (omotracheal)	Occipital · supraclavicular (omoclavicular, subclavian)

TRIANGLES OF THE NECK — schematic side view (face to the left)**2 Anterior triangle — the four subdivisions**

Triangle	Borders	Contents (per slides)
Submental	Below: hyoid · lateral: anterior belly of digastric · medial: midline (floor: mylohyoid)	Anterior jugular veins, submental lymph nodes
Submandibular (digastric)	Above: inferior border of mandible · anterior and posterior bellies of digastric	Submandibular gland and nodes, caudal parotid · facial artery and vein, submental and lingual vessels · mylohyoid and hypoglossal (XII) nerves
Carotid	Front: superior belly of omohyoid · above: stylohyoid + posterior belly of digastric · behind: anterior border of SCM	Common, external, internal carotid (and sinus); ECA branches except maxillary, superficial temporal, posterior auricular · IJV, common facial, lingual, superior and middle thyroid veins · vagus (X), hypoglossal (XII) , sympathetic trunk
Muscular (omotracheal)	Above: hyoid · lateral: superior belly of omohyoid + anterior border of SCM · medial: midline	Thyroid, sternothyroid, sternohyoid · superior and inferior thyroid arteries, anterior jugular veins · thyroid, parathyroids, larynx, trachea, esophagus

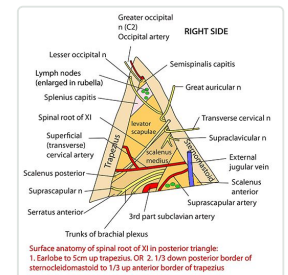
Whole anterior triangle	Muscles: infrahyoids · organs: thyroid, parathyroids, larynx, trachea, esophagus, submandibular gland, lower parotid · arteries: superior/inferior thyroid, carotids, facial, submental, lingual · nerves: X, XII , sympathetic trunk, mylohyoid nerve
Carotid sheath (slide figure)	Encloses common/internal carotid, IJV and vagus · CN IX, X and XI emerge near its upper end · superior laryngeal nerve → internal and external laryngeal branches



Carotid triangle and its contents.

3 Posterior triangle

Shape	Spiral (twists round the neck)
Roof	Investing fascia, platysma, external jugular vein
Floor	Prevertebral fascia over semispinalis, splenius, levator scapulae, scalenus medius (and posterior) · subclavian artery, trunks of brachial plexus, cervical plexus
Arteries	Occipital, superficial (transverse) cervical, suprascapular · 3rd part of subclavian
Veins	Transverse cervical, suprascapular, external jugular
Nerves	Cervical plexus branches (lesser occipital, great auricular, transverse cervical, supraclavicular) · spinal root of accessory (XI) · brachial plexus trunks · suprascapular nerve
Muscle	Inferior belly of omohyoid with its sling → splits occipital (above) from supraclavicular (below) triangles
Nodes	Occipital (enlarged in rubella and scalp infections) · supraclavicular (part of the deep chain)

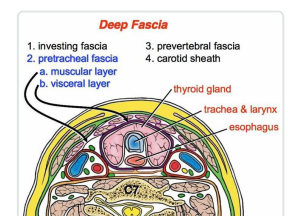


Contents of the posterior triangle (right side).

Surface marking of the accessory nerve (XI) in the posterior triangle (slide): earlobe → a point 5 cm up the trapezius, or from 1/3 of the way down the posterior border of SCM to 1/3 of the way up the anterior border of trapezius.

4 Deep cervical fascia

Layer	Key facts
Superficial fascia	Contains platysma , anterior and external jugular veins and cutaneous nerves
1 Investing	Encircles the neck · splits to enclose SCM and trapezius and the parotid ; superficial layer = parotid fascia, attached to the zygomatic arch · attachments: mandible, mastoid, external occipital protuberance and superior nuchal line, ligamentum nuchae, acromion and spine, clavicle, manubrium (suprasternal space of Burns)
Pierced by	External and anterior jugular veins · lesser occipital, great auricular, transverse cervical, supraclavicular nerves (all cutaneous branches of the cervical plexus)
2 Pretracheal	Muscular layer (infrahyoids) + visceral layer around thyroid, trachea/larynx, esophagus · runs down to the arch of the aorta
3 Prevertebral	In front of the vertebral column and its muscles (longus colli) · forms the floor of the posterior triangle · prolonged over the subclavian vessels and brachial plexus as the axillary sheath · retropharyngeal space lies in front of it, behind the buccopharyngeal fascia, down to ~T3
4 Carotid sheath	Around the carotid artery, IJV and vagus



Layers of deep cervical fascia (C7 section).

Compartments in cross-section (slide figure): **visceral** (pretracheal), **vascular** (carotid sheaths), **vertebral** (prevertebral).

5 Muscles

Muscle	Origin → insertion	Nerve	Action
Sternocleidomastoid	Manubrium + medial clavicle → mastoid process	(spinal accessory)*	One side: tilts head to same side, turns face to opposite side · both: draw head back
Trapezius	EOP, superior nuchal line, spines C7–T12 → lateral ⅓ clavicle, acromion, scapular spine	(spinal accessory)*	Elevates, adducts, depresses scapula
Digastric	Anterior belly: digastric fossa of mandible · posterior belly: mastoid notch → intermediate tendon to hyoid	Ant: nerve to mylohyoid (V3) · post: facial (VII)	Opens mouth; raises hyoid
Stylohyoid	Styloid process → body/greater horn of hyoid	Facial (VII)	Pulls hyoid up and back
Mylohyoid	Mylohyoid line → hyoid + median raphe	Nerve to mylohyoid (V3)	Supports and raises floor of mouth
Geniohyoid	Inferior genial tubercle → hyoid	C1 via hypoglossal	Elevates hyoid / depresses mandible
Sternohyoid	Manubrium + medial clavicle → hyoid	Ansa cervicalis (C1–C3)	Depresses hyoid after swallowing
Omohyoid	Upper border of scapula near suprascapular notch → hyoid	Ansa cervicalis	Depresses and fixes hyoid
Thyrohyoid	Oblique line of thyroid cartilage → hyoid	C1 via hypoglossal	Depresses hyoid / raises larynx
Sternothyroid	Manubrium → oblique line of thyroid cartilage	Ansa cervicalis	Draws larynx down

*The slides give no nerve for SCM and trapezius; the spinal accessory nerve is added from standard anatomy. Root of the neck (slide figure): **phrenic nerve (C3–5)** on scalenus anterior; brachial plexus roots/trunks and subclavian artery pass through the **interscalene triangle**, subclavian vein in front of scalenus anterior.

P Must-remember pearls

- Anterior triangle: mandible, midline, anterior border of SCM · posterior: SCM, trapezius, middle ⅓ clavicle.
- Anterior = submental + submandibular + carotid + muscular · posterior = occipital + supraclavicular (split by inferior omohyoid).
- Submental: hyoid, anterior digastric, midline → submental nodes, anterior jugular veins.
- Submandibular: mandible + two digastric bellies → submandibular gland, facial vessels, XII, mylohyoid nerve.
- Carotid: superior omohyoid, posterior digastric + stylohyoid, SCM → carotids, IJV, X, XII, sympathetic trunk.
- Posterior triangle roof = investing fascia + EJV · floor = prevertebral fascia · key nerve = spinal accessory (XI).
- Investing fascia encloses SCM, trapezius and parotid; pierced by EJV, AJV and the cutaneous cervical plexus nerves.
- Pretracheal = infrahyoids + thyroid, trachea, esophagus · prevertebral → axillary sheath, floor of posterior triangle.
- Digastric: anterior belly V3, posterior belly VII · stylohyoid VII · geniohyoid and thyrohyoid C1 via XII · other infrahyoids ansa cervicalis.
- SCM: turns the face to the **opposite** side.

Q Test yourself

1. During a lymph node biopsy in the posterior triangle, the patient later cannot shrug his shoulder. Which nerve was injured?

- A. Phrenic nerve
- B. Great auricular nerve
- C. Spinal accessory nerve (XI)
- D. Hypoglossal nerve
- E. Suprascapular nerve

2. A surgeon opens the triangle bounded by the superior belly of omohyoid, the posterior belly of digastric and the anterior border of SCM. Which nerve lies here?

- | | |
|---------------------------|--------------------|
| A. Vagus nerve | B. Phrenic nerve |
| C. Lesser occipital nerve | D. Mylohyoid nerve |
| E. Suprascapular nerve | |

3. A patient has an infection of the chin skin and lower lip tip. Lymph nodes in which triangle enlarge first?

- | | |
|--------------------|--------------|
| A. Carotid | B. Muscular |
| C. Occipital | D. Submental |
| E. Supraclavicular | |

4. Which muscle divides the posterior triangle into occipital and supraclavicular parts?

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|---------------------------------|-------------------------------|
| A. Posterior belly of digastric | B. Inferior belly of omohyoid |
| C. Scalenus anterior | D. Stylohyoid |
| E. Sternohyoid | |

5. Contraction of the right sternocleidomastoid alone turns the face to which side?

- | | |
|--------------------------|-----------------------------|
| A. Right, chin raised | B. Straight forward |
| C. Down toward the chest | D. Neither — it only flexes |
| E. Left (opposite side) | |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 C** — The spinal root of the accessory nerve crosses the posterior triangle superficially and supplies trapezius.
- 2 A** — This is the carotid triangle: carotids, IJV, vagus, hypoglossal and sympathetic trunk.
- 3 D** — The submental triangle contains the submental lymph nodes.
- 4 B** — The inferior belly of omohyoid crosses the posterior triangle, separating occipital from supraclavicular triangles.
- 5 E** — One SCM tilts the head to its own side and turns the face to the opposite side.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy neck triangles slides, 2026-2027
Study alongside the lecture, not instead of it.